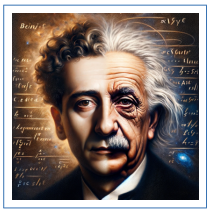
Xianquan Yan

Curriculum Vitae

Department of {Physics, Computer Science}
National University of Singapore
Homepage *Github* *Linkedin*



Education

2023–present **PhD Candidate, {Physics, Computer Science}**, *National University of Singapore*, Singapore.
Supervisors: Lee Ching Hua and Kenji Kawaguchi
Research Interests: Topological Non-Hermitian Physics, AI + Science, Complex Systems, Quantum Information Science
GPA: *5.00/5.00*

2019–2023 **BS, Physics**, *Sichuan University*, Chengdu, China.
GPA: *3.96/4.00* (Ranking 1/285)

2022 **Visiting Student (BPIE), Physics**, *University of California, Berkeley*, CA, USA, *GPA: 4.00/4.00*.

2022 **Summer Institute, Physics**, *University of Oxford*, Oxford, UK, *GPA: 90/100*.

Publications

**Equal contribution. #Corresponding author.*

2026 Xianquan Yan, Hakan Akgün, Kenji Kawaguchi, N. Duane Loh, and Ching Hua Lee. "Hsg-12m: A large-scale benchmark of spatial multigraphs from the energy spectra of non-hermitian crystals". In **International Conference on Learning Representations (ICLR) 2026**, Jan 2026.

2026 Zhaoyun Chen*, Xianquan Yan*, Hakan Akgün, N Duane Loh, and Ching Hua Lee. "Multi-modal classification of extended 1d cellular automata". In **AI4X — Accelerate Conference 2026**, Mar 2026.

2026 Hakan Akgün, Xianquan Yan, Tamer Taşkıran, Muhamet Ibrahimi, Ching Hua Lee, and Seymur Jahangirov. "Deterministic scale-invariant dynamics in a logistic game-of-life model". **Communications Physics, Nature Publishing Group UK London**, Mar 2026.

2025 Xianquan Yan and Jian-Song Pan. "Precise identification of topological phase transitions with eigensystem-based clustering". In **AI4X 2025 International Conference**, Apr 2025.

2025 Xianquan Yan, Hakan Akgün, Kenji Kawaguchi, N Duane Loh, and Ching Hua Lee. "Classifying the graph topology of non-hermitian energy spectra with graph transformer". In **AAAI 2026 XAI4Science Workshop**, Nov 2025.

2025 Xianquan Yan#. "Automated metric discovery: Navigating quantum geometry with symbolic regression". In **AI4X 2025 International Conference**, Apr 2025.

Experience

2024–2026 **Teaching Assistant**, *National University of Singapore*, Singapore.
- (PC5253) Complex Systems Analysis and Modelling
- (AIS5102) Practical Machine Learning for Scientific Discovery
- (AIS5204) AI in Condensed Matter Physics

2022 **Student Research Assistant**, *Lawrence Berkeley National Laboratory*, Berkeley, CA, USA.
Supervisor: Benjamin Nachman

Academic Service

2026 Reviewer, *Communications Physics*, Nature Publishing Group UK London

Codebases and Datasets

`poly2graph`: Python package for automatic Hamiltonian spectral graph construction from characteristic polynomials and Bloch Hamiltonians.
[https://github.com/sarinstein-yan/poly2graph]

`HSG-12M`: Codebase for downloading, processing, subsetting, and benchmarking the HSG-12M spatial multigraph dataset with NetworkX and PyTorch Geometric.
[https://github.com/sarinstein-yan/HSG-12M]

HSG-12M (Data): Harvard Dataverse dataset of spatial multigraphs extracted from the energy spectra of non-Hermitian crystals.
[https://doi.org/10.7910/DVN/PYDSSQ]

`EigsysClustering`: Open-source toolkit for unsupervised identification of topological phases using *eigensystem vectors* and Gaussian mixture model clustering.
[https://github.com/sarinstein-yan/EigsysClustering]

`ClusterAnalyzer`: Python library for analyzing critical behavior, self-similarity, and fractal dimensions of clusters in 2D cellular automata.
[https://github.com/HakanAkgn/ClusterAnalyzer]

Honors and Awards

2021 **China National Scholarship** (Top 0.2%), The Ministry of Education of the P.R. China

2020 **China National Scholarship** (Top 0.2%), The Ministry of Education of the P.R. China

2022 **1st Prize, Scholarship of *Bajian* program for fundamental disciplines 2.0**, The Ministry of Education of the P.R. China

2022 **1st Prize, CASC Scholarship**, China Aerospace Science and Technology Corp

2012 **Gold, Asia International Mathematical Olympiad Open Contest**, A.I.M.O. Union

2019 **Provincial Excellent Student Leader**, The Education Department of Sichuan Province

2020 **1st Prize** and **Best Demeanor Award**, South-West Undergraduate Physics Tournament, Sichuan Physical Society

2020 **2nd Prize, National English Competition for College Students**, Advisory Board for College Foreign Language Teaching

2026 **Faculty of Science PhD Conference Award (FPCA)**, Faculty of Science, National University of Singapore

Extracurricular Activities

2013–2023 **Football (Soccer)**, *Captain, Left-Winger*, School Team.

2019–2021 **Debater**, *Debate Team (English) of Sichuan University*.

2019–2021 **Debater**, *Debate Team (Chinese) of College of Physics, Sichuan University*.

2021–2023 **Director**, *Sichuan University Mathematica Society*.

2019–2021 **Director**, *Sichuan University English Association*.

Calligraphy.

Marathon Runner.